

Exploring the types of data on machine learning.

Although I have gotten familiar with machine learning since high school, before taking the lab, I initially thought that the machine learning workflow was a series of separate technical steps. After taking a look through the lab, I was able to combine what I've learned about machine learning in high school and the new topics from this lab and later understood that in machine learning, each of the stages build logically compared to the previous. The steps to this process begin with understanding the data and therefore preparing it, before undergoing a series of operations which includes feature selection, data splitting, model training, and evaluation, and interpretation of the final results based on the provided source data. All of this goes to show that the data quality provided to the systems including the variables, and the type of evaluation method used all have a form on influence to the final results, proving that they are important as the algorithm itself.

In the context of data types used for interpretation, supervised learning utilized labeled examples which interprets the system to define the variables as well as the correct answer in the training process which enhances class prediction. Unsupervised learning is the opposite to this, as there is no defined target label that the system uses in the training process and therefore no answer key, which is useful for identifying various groups, what is similar, and what patterns it can see. Reinforced learning is different and this is because the system learns through trial and error as evident by a reward-penalty system.

In the original logistic regression model system, the model achieved an accuracy of 88.9%, while the decision tree had an 83.3% accuracy by approximation. The fact that the logistic regression model system had higher accuracies proves that a much flexible model had better results. The lower accuracy score from the decision tree can be attributed to the maximum depth of three, which created less accurate results due to its simplicity. These results from the experiment showcases the fact that inputs and impact and affect the performance which illustrates a clear picture on how feature selection, cross-validation, and repeated testing all play a critical step in machine learning and how a single result is not enough without considering how the data was categorized and refined.

Energy and engineering operations can be a beneficiary to this because it helps make predictions and therefore get the data provided by the customer and client, as well as the cost, budgeting, and economic region among other things/ It enables that model to help identify management risks in the process while also ensuring that the project meets the criteria. In the context of this, it helps reduce the time and tasks it takes to get projects done while also enabling the project to meet its satisfied goals.

In conclusion, the most important takeaway from the lab was the fact that machine learning is built on the basis of end-to-end decision process instead of simply running an algorithm. What's challenging about this is the fact that performance changes can affect the settings of the models and features of the machine learning system. Some concepts that I want to learn next include feature scaling, cross-validation, hyperparameter tuning, and additional models among others to explore the machine learning subcategories in further detail. I also want to know how all of this can influence various metrics when there are irregularities in the subclasses.